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RESEARCH ARTICLE

A Decision-Making Model for Autonomous Vehicles Considering Pedestrian's Time Pressure Based on Game Theory and Reinforcement Learning

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This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

ABSTRACT Due to the obvious randomness, pedestrian crossing behavior is hard to predict, which challenges the decision-making of autonomous vehicles (AVs). Recent solutions have been able to adapt to structured road scenes with crossing signals or markings. However, there is still a gap in extending the pedestrian-vehicle interaction (PVI) performance in structured road scenes to unstructured road scenes. Therefore, this paper proposed a vehicle decision-making model considering pedestrian intention based on game theory and reinforcement learning (RL). We designed and conducted a simulation experiment based on a virtual reality platform. Then, leveraging game theory, we established a pedestrian crossing decision-making model considering pedestrian heterogeneity evoked by time pressure (TP). A reward function was developed to enhance driving performance by combining safety, efficiency, and comfort. The RL agent of AVs learns to control the vehicle speed in a pattern that maximizes cumulative rewards through trials and errors by interacting with pedestrians in the simulation environment. The results show that AVs can effectively and safely interact with heterogeneous pedestrians on unstructured roads based on the proposed model. This study contributes to developing AVs that interact better with pedestrians and improve traffic safety, efficiency, and user acceptance of autonomous vehicles.

INDEX TERMS Autonomous vehicle, decision-making, pedestrian-vehicle interaction, time pressure, unstructured roads.

I. INTRODUCTION

Autonomous vehicles (AVs) hold the promise of transforming urban transport by enhancing traffic safety, reducing emissions, and optimizing traffic flow for greater efficiency [1]. Central to realizing these benefits is the capability of AVs to interact efficiently and safely with pedestrians and other traffic participants [2]. However, the attention and behavior of pedestrians are affected by various factors such as age, road conditions, weather and so on, leading to uncertainty and unpredictability [3]. Therefore, interacting with pedestrians is a key challenge, especially on unstructured roads with complex pedestrian crossing behavior and frequent traffic participants' competition for the right of way. An unstructured road section refers to a road that lacks traffic markings and controls that are not in place [4]. In this

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paper, it is limited to a road that is not marked with vehicle and pedestrian signals and pedestrian crossing markings. Compared with structured roads, there are more frequent pedestrian-vehicle interactions (PVI) in unstructured road sections. In that case, AVs are more likely to face the problem of freezing robots, which means pedestrians could abuse their safety strategies due to an over-conservative system [5]. However, there is currently a lack of mature schemes to solve the modeling and control problem of daily interaction between AVs and pedestrians in a specific environment. To improve the safety, efficiency and social acceptance of autonomous vehicles, researchers must design a powerful decision-making system to deal with uncertain pedestrian behavior.

Autonomous driving decision-making models oriented to pedestrian-vehicle interaction (PVI) require vehicles to predict pedestrian crossing intention and make suitable decisions. In recent years, research on the decision-making in the progress of PVI has experienced a substantial boost, such as game theory, Markov sequence decision progress, etc. Game theory is widely applied to model decision-making for the interactions between pedestrians and vehicles. Researchers conducted classic game theory models, studied the utility function of each game participant, solved the model, analyzed the Nash equilibrium state and selected the optimal strategy for the game process of pedestrians and vehicles. For example, the Pedestrian-Vehicle Game (PVG) model was used to analyze the uncertainty in the dynamic change process between "pedestrian crossing first" and "vehicle crossing first", and obtain the optimal strategy based on the Nash equilibrium [6]. On this basis, the extended PVG model considered the time pressure (TP) and other factors that pedestrians may face when crossing the street, improved the benefits in the game process, and solved the mixed advantage strategy combination under specific conditions [7]. In addition, the cumulative prospect theory was used to solve the conflict between game theory and bounded rationality, better simulate the variability of human behavior in the real world and provide a reference for real-time decision-making of AVs [8], [9].

For the decision-making and control of AVs, reinforcement learning (RL) allows agents to get the optimal strategy by interacting with the environment [10]. Among them, the hidden Markov model (HMM), partially observable Markov decision process (POMDP) and mixed observable Markov decision processes (MOMDP) are widely used to model the decision-making of AVs. In POMDP or MOMDP, the agents of AVs learned to capture the uncertainty of pedestrians implied in the environment and make relatively optimal decisions through trial and error [11], [12], [13]. In other research, HMM was used to predict future states and estimate the likelihood that humans will cross the path, and concurrently learn, update, and improve the predictive model with new observations at run-time, so as to ensure active motion planning control safely [14].

The existing research on PVI vehicle decision-making can predict pedestrian intentions in most cases. However, for certain scenarios like unstructured roads, pedestrian motion data is still scarce. Moreover, although studies on modeling pedestrian-vehicle interaction have considered pedestrian heterogeneity, there has been no research on vehicle decision-making facing pedestrians with TP.

The objective of this paper is to explore how autonomous vehicles make speed decisions when facing heterogeneous pedestrians on unstructured road scenes. We have expanded the dataset in such a scenario and proposed a vehicle speed decision model based on game theory and deep reinforcement learning, which enables effective interaction between vehicles and heterogeneous pedestrians on unstructured roads.

The main contributions of this study are summarized as follows:

(1) We designed a pedestrian crossing simulation experiment in VR and extracted the pedestrian crossing data considering heterogeneity evoked by TP.
(2) Based on game theory, we established a pedestrian crossing decision-making model considering pedestrian heterogeneity evoked by TP, vehicle velocity, and pedestrian-vehicle distance on the unstructured road.
(3) We developed a RL-based framework for AVs to obtain more reasonable and effective speed decisions, considering the uncertainty of pedestrian crossing intentions on unstructured roads.

The rest of this paper is organized as follows. Section II introduces the pedestrian crossing simulation experiment and data preparation. Section III shows the model of the pedestrian crossing decision based on game theory. Section IV shows the RL framework and experimental results of training AV speed decisions. Finally, Section V ends this study with conclusions and future work.

II. HUMAN IN THE LOOP EXPERIMENT

A. SCENARIO AND APPARATUS

We conducted a pedestrian crossing simulation experiment to explore the uncertainty of pedestrian crossing decisions on unstructured roads. The unstructured road in this study is designed as a lane without signal control and crossing marks. The scenario is shown in Figure 1, in which subjects select whether to cross the road in a virtual scene of an unstructured road facing two vehicles with a length, width, and height of 4.56 m, 1.76 m, and 2.02 m, respectively passing in sequence at a certain gap.

A pedestrian simulator, composed of simulation software (i.e., UC win/road 15.1), VR glasses (i.e., Oculus Rifts), and 2 IMU sensors, was used to simulate the walking motion of the subjects in the virtual scene, as shown in Figure 2.

B. PARTICIPANTS

A total of 54 subjects were recruited, including 40 males and 14 females with a mean age of 21.83 years

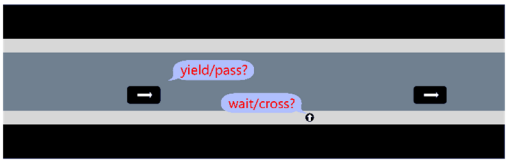


FIGURE 1. HITL experiment scenario.

(standard deviation of 1.40 years) and an age range of 19-24 years old. All subjects reported normal vision or corrected vision and normal hearing. The subjects received an honorarium at the end of the experiment.

C. EXPERIMENTAL DESIGN

The experiment followed a $2 \times 2 \times 9$ within-subjects design. The independent variables include time pressure (with or without TP), driving decision of the vehicle (yielding to a stop or passing at a constant speed), and vehicle gap (2 s-6 s, with an interval of 0.5 s).

The vehicle gap refers to the time headway between the rear end of the front vehicle and the front end of the rear vehicle. For the TP, the subjects were told before the experiment, "Imagine that your good friend is already on the other side of the road, and he/she is about to leave, and you need to speak to him/her right away about something important, and you are worried that if you don't get across soon, he/she will leave without you. So, you want to cross the street as fast as you can, but remember that you can only cross the street when you feel it is safe" [15]. At the same time, a visual alert in the form of stopwatches appeared in the virtual scene, as shown in Figure 3. The driving status of the vehicle includes decelerating to a stop and passing at a constant speed of 50 km/h.

The dependent variable is gap acceptance behavior. Gap acceptance behavior can be quantitatively depicted through the probability of choosing to cross the street at varying intervals between approaching vehicles. The vehicle gap when a pedestrian chooses to cross is called a gap acceptance. Conversely, it is called a rejected gap.

At the beginning of the experiment, two vehicles that maintained a certain distance between them were traveling at a constant speed of 50 km/h, and the front vehicle was 69.45 meters away from the pedestrian. The vehicle was traveling

in the middle of the road. Note that the subjects were only allowed to cross the street at the gap between the two vehicles or after the rear vehicle had driven by. Each subject was required to complete 36 trials, and we used a Latin square design to balance the order of the trials to avoid order effects. The total duration of the experiment was about 1.5 hours.

D. PROCEDURE

Upon arrival at the lab, the participants need to fill in a demographic questionnaire. The experimenter then explained the experiment content to the subjects until they fully understood it. Then, the subjects wore the VR headset and sensors and practiced walking in the virtual environment until they were familiar with the walking operation and crossing task. Before the formal experiment, the subjects were instructed to mimic actual crossing behavior and prioritize safety. When the TP was present, the subjects were required to complete the crossing task under the TP condition. After the experiment, the subjects were asked to complete a post-test questionnaire and an interview.

E. DATA DESCRIPTION AND ANALYSIS

A total of 1944 trials were conducted. The original data for each set of trials consisted of a sequence of trajectories (position and speed) of pedestrians and vehicles. Figure 4 shows the statistical results of pedestrians' gap acceptance behavior under constant speed conditions. The probability of pedestrians choosing to cross increases with the increase of vehicle gap. In addition, the pedestrian crossing probability with TP is significantly higher than that without TP.

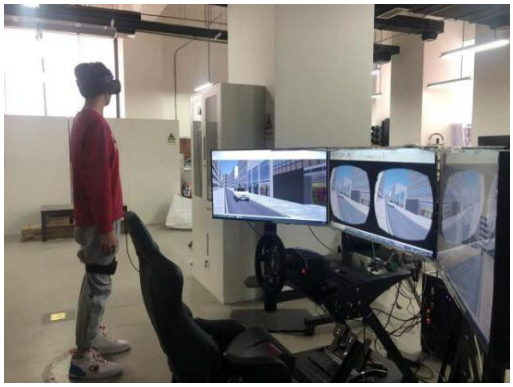
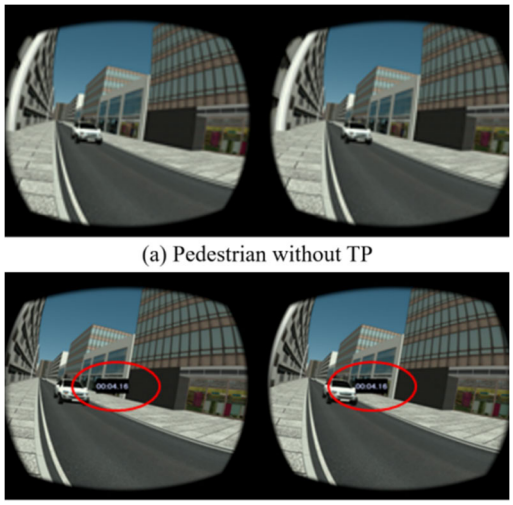


FIGURE 2. Pedestrian simulator.



(a) Pedestrian without TP

(b) Pedestrian with TP

FIGURE 3. Pedestrian-vehicle interaction scene from the perspective of a pedestrian.

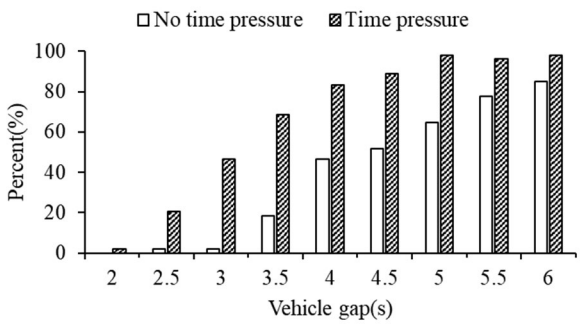


FIGURE 4. Statistics on pedestrian gap acceptance behavior.

III. PEDESTRIAN CROSSING MODEL BASED ON GAME THEORY

A. GAME PAYOFF

We use an extended PVG model to estimate the uncertainty of pedestrian crossing decisions. The PVG model includes three components: players, strategies, and payoff. The two players are the pedestrian and the vehicle (i = 1,2), and the set of strategies includes crossing and waiting. The model decomposes the payoff into the weighted sum of the following utilities:

(1) The unpleasant experience resulting from a collision between two players is defined as the utility of risk perception: k = 1/TTC;
(2) The time loss incurred by one player choosing to yield is defined as the utility of delay, with a value equal to the time taken by the other player to pass the crossing position ti.

Assume that these two utilities always exist in all situations. Table 1 shows the parameters of the PVG model.

TABLE 1. Parameters description of the PVG model.

Parameters	Description
v	Vehicle speed (m/s)
l	Distance between vehicles and pedestrians (m)
k	The utility of risk perception for pedestrians or vehicles
t	The utility of delay for pedestrians or vehicles
t_1	Time required for the pedestrian to pass through the crossing location (s)
t_2	Time required for the vehicle to pass through the crossing location (s)
c	Multiplier for the utility of delay when both the pedestrian and vehicle choose to choose
a	A factor reflects crossing intention varying with the cumulative waiting time of pedestrians

The vehicle or pedestrian game payoffs were designed to satisfy the following principles:

(1) If both the vehicle and pedestrian choose to cross, they will lose the utility of risk perception and the utility of delay.
(2) When the pedestrian chooses to cross while the vehicle chooses to wait, the pedestrian will gain the utility of delay and the utility of risk perception, while the vehicle will gain the utility of risk perception, but lose the utility of delay due to the choice to wait, with the same utility setting in the opposite case.
(3) When both the vehicle and the pedestrian choose to wait, they will both gain the utility of risk perception but lose the utility of delay at the same time due to the choice to wait.

The game payoff matrix is shown in Table 2, where pedestrian crossing/waiting is abbreviated as PC/PW and vehicle crossing/waiting is abbreviated as VC/VW.

TABLE 2. Game payoff matrix.

	Pedestrian cross		Pedestrian wait	
Vehicle cross	$-k_1 - at_1$	$-k_2 - at_2$	$k_1 + at_1$	$k_2 - at_2$
Vehicle wait	$k_1 - at_1$	$k_2 + at_2$	$k_1 - at_1$	$k_2 - at_2$

B. MODEL DESIGN

We used the mixed strategy algorithm to solve the game model. This algorithm assumes that the pedestrian adopts mixed strategies $U_P = (P_{PC}, 1\text{-}P_{PC})$ while the vehicle adopts mixed strategies $U_V = (P_{VC}, 1\text{-}P_{VC})$. According to the principle of payoff maximization, the expected utility function of the vehicle is as follows:

$$V_V\left(U_V, U_P\right) = P_{VC}\left(P_{PC}V_{C,C}^V + (1-P_{PC})\,V_{C,W}^V\right) + (1-P_{VC})\left(P_{PC}V_{W,C}^V + (1-P_{PC})\,V_{W,W}^V\right) \tag{1}$$

To maximize the utility of the pedestrian cross, such that $\delta V_V/\delta P_{VC} = 0$, there are $P_{PC}VV\ C,C + (1\text{-}P_{PC})\ VV\ C,W - P_{PC}VV\ W,C - (1\text{-}P_{PC})\ VV\ W,W = 0$. Thus, the model can be organized as follows:

$$P_{PC} = \frac{V_{C,W}^V - V_{W,W}^V}{V_{C,W}^V + V_{W,C}^V - V_{C,C}^V - V_{W,W}^V} \tag{2}$$

where P_{PC} is the probability that a pedestrian chooses to cross the street, $VV\ C,W$ is the payoff of the vehicle when the vehicle chooses to cross the section and the pedestrian chooses to wait, $VV\ W,W$ is the payoff of the vehicle when both the vehicle and the pedestrian choose to wait, $VV\ W,C$ is the payoff of the vehicle when the vehicle chooses to wait and the pedestrian chooses to cross, and $VV\ C,C$ is the payoff of the vehicle when both the vehicle and the pedestrian choose to cross.

Therefore, according to Equation (2), the pedestrian crossing decision-making model based on the PVG model is defined as follows:

$$P_{PC} = \frac{2at_1}{2k_1 + (1+c)\,at_1} \tag{3}$$

where PPC is the probability that a pedestrian chooses to cross.

The PVG model considers the current pedestrian-vehicle distance and vehicle speed, which implies the strategy of vehicles in the game-theoretical model. The extended PVG model further considers TP as a factor affecting pedestrian decision-making in this paper and introduces a stochastic factor to solve the bounded rationality in the pedestrian crossing decision-making process and the variability of pedestrian behavior in the real world. The added variables and parameters of the model are shown in Table 3. In the extended PVG model, the probability of a pedestrian choosing to cross PPC is expressed as follows:

$$P_{PC} = \frac{l\left(l+l'\right)\left[2a_1 + 2\left(a_2 - a_1\right)T_p\right]}{2v^2 + l\left(l+l'\right)\left[\left(1+c_2\right)a_2T_p + \left(1+c_1\right)a_1\left(1-T_p\right)\right]} + \xi \tag{4}$$

TABLE 3. New parameters in the extended PVG model.

Parameters	Description
l'	Vehicle length (m)
Tp	Time pressure (0/1, without/with TP)
c_1, c_2	Multiplier for the utility of delay without/with TP (emergency braking)
a_1, a_2	Weighting coefficients for without/with TP
ξ	The stochastic variables of the decision-making process

TABLE 4. Errors of pedestrian crossing decision model.

	RMSE	MAE	R^2
With TP	6.53%	0.43%	0.963
Without TP	6.60%	0.44%	0.958

C. PARAMETER SOLUTION

Based on the data obtained under different TP and gap conditions, the Adam algorithm is selected to calibrate the parameters of the proposed pedestrian crossing decision model. The advantage of the Adam algorithm is that it can dynamically adjust the learning rate of each parameter and solve the problem of constant step size in the SGD algorithm [16]. The calculation results of the parameter calibration are as follows: parameters ξ, a1, c1, a2, c2 are −0.4842, 0.0946, 0.0011, 0.1807, and 0.0004, respectively.

The pedestrian crossing decision-making model considering TP is shown as follows:

$$P_{PC} = \frac{l\,(l+4.57)\left(0.189 + 0.172T_p\right)}{2v^2 + l\,(l+4.57)\left(0.086T_p + 0.095\right)} - 0.484 \tag{5}$$

In addition, the output value of Equation (5) is limited between 0 and 1 by clamping as an estimate of pedestrian crossing probability. We selected the root mean square error (RMSE), mean absolute error (MAE), and coefficient of determination (R2) as the evaluation metrics for the proposed model. The error statistics are shown in Table 4. The pedestrian crossing probability is shown in Figure 5, where pedestrians with TP have a higher crossing probability than pedestrians without TP. We found that the pedestrian crossing decision model proposed was effective in predicting pedestrian crossing decisions under different TP conditions.

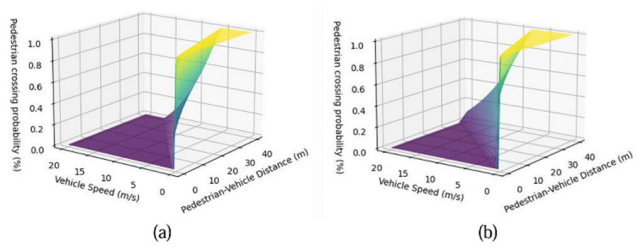


FIGURE 5. Pedestrian Crossing Probability (a)without TP (b)with TP.

IV. PEDESTRIAN CROSSING MODEL BASED ON GAME THEORY

Since vehicle acceleration is a continuous variable, and it will be a sparse event for the pedestrian to choose to cross the street and happen to collide with the vehicle in the simulation, the Soft Actor-Critic (SAC) algorithm was used to solve the MDP [17]. In this section, pedestrian crossing decisions were modeled as the hidden state, and the RL agent proposed to learn the speed control strategy using SAC is explained.

A. STATE AND ACTION DEFINITION

At a certain time step t, the vehicle state of a PVI process is described by the AV speed v_t and pedestrian-vehicle distance d_t. The action is the longitudinal acceleration of the AV a_t. In this study, the action set (longitudinal acceleration selection set) is continuous, ranging from −7.5 m/s^2 to 5 m/s^2. Given state and action at time step t, the next-step state is updated by a kinematic point-mass model:

$$\begin{cases} d_{t+1} = d_t - v_t\Delta t - \frac{1}{2}a_t\Delta t^2 \\ v_{t+1} = v_t + a_t\Delta t \end{cases} \tag{6}$$

where delta is the simulation time interval, set as 0.1 s in this study.

At a certain time step t, the pedestrian state of a PVI process is described by crosswise position x_t, crossing speed v_p and crossing probability p of pedestrians. Assume that in each time step t, pedestrians on the roadside choose to be stationary or cross the street with p. After pedestrians choose to cross the street, the pedestrian crossing intention and speed will remain unchanged throughout the planning cycle, and the state transfer will be carried out according to the following formula:

$$x_{t+1} = x_t + v_t \Delta t \tag{7}$$

B. SIMULATION SETUP

To encourage the RL agent to learn from trial and error, we developed a PVI simulation environment in Carla, as shown in Figure 6.

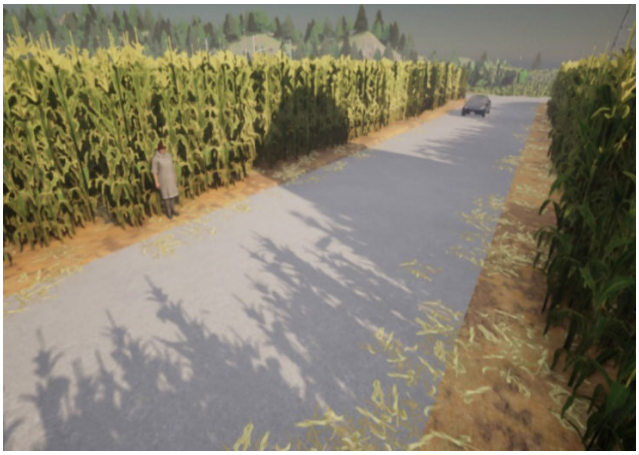


FIGURE 6. Pedestrian-vehicle interaction simulation environment.

The simulation involves two agents: the pedestrian agent (PA) and the vehicle agent (VA). PA street crossing behavior follows the modeling based on experiment data, wherein pedestrian crossing speed with TP and without TP follows a skewed normal distribution with parameters (−2.25, 1.78, 0.36) and (−2.06, 1.77, 0.35), respectively, and pedestrian crossing probability is calculated by Equation (5). Meanwhile, the VA is controlled by the SAC algorithm.

During initialization, specify the pedestrians who are stationary on the roadside and may cross the street, the vehicle position (uniformed between 20 m and 40 m away from pedestrians) and the initial speed of 40 km/h. Given acceleration, future VA velocity, and pedestrian-vehicle distance are then generated iteratively based on Equation (6). Facing VA, PA still on the roadside will cross the street selectively. At each time step, the simulation environment provides the RL agent with a reward value (calculated based on time progress, speed, safety, and jerk) to the RL agent as feedback to make the RL agent learn a reasonable strategy. In addition, assume that after the pedestrian makes the crossing decision, set the pedestrian crossing probability to 1, and the subsequent crossing speed will not change. After a complete crossing, set the crossing probability and crossing speed to 0.

C. REWARD FUNCTION

The reward function, r(s,a), serves as a training signal to encourage or discourage behaviors in the context of a desired task. For the task of AV velocity decision, a reward function was established based on a linear combination of the features:

$$r = w_1 R_{col} + w_2 R_{speed} + w_3 R_{jerk} + w_4 R_{pass} \tag{8}$$

where w1, w2, w3, and w4 are positive weighting scalars corresponding to collision evaluation R_{col}, stable-speed evaluation R_{speed}, comfort evaluation R_{jerk}, and time evaluation R_{pass}, respectively. The four features are defined as follows:

(1) Collision evaluation: The collision evaluation R_{col} is set to $-(v^2+0.5)$ if there is an accident risk of collision with vehicles during the pedestrian crossing, otherwise $R_{col} = 0$. Set accident risk conditions to a speed greater than 1 m/s and a pedestrian-vehicle distance of less than 1.5 m.

(2) Stable-speed evaluation: To encourage the vehicle to pass at a higher but not too excessive speed, the stable-speed evaluation R_{speed} is defined as follows:

$$R_{speed} = -|v - v_{\max}| / v_{\max} \tag{9}$$

(3) Comfort evaluation: The comfort evaluation is defined as follows:

$$R_{jerk} = -jerk^2 \tag{10}$$

where Jerk, the change in acceleration, was considered as a measure of driving comfort [18].

(4) Time evaluation: To encourage vehicles to maximize the comprehensive benefits of a PVI event, the time evaluation R_{pass} is set to $1/(T_{car} + T_{ped})$ after both the vehicle successfully passes the pedestrian and the pedestrian crosses the street, otherwise, $R_{pass} = 0$.

In the time evaluation R_{pass}, T_{car} refers to the time it takes for the vehicle to travel from the starting point to a certain distance beyond the pedestrian (10 m in this study), and T_{ped} refers to the time it takes for the pedestrian to completely cross the street from the starting point.

D. SOFT ACTOR-CRITIC

As an actor-critic and model-free algorithm, SAC introduces an additional term, maximum entropy, on top of the traditional RL algorithm's goal to train an agent to maximize the expected sum of rewards [19].

$$\sum_{t=0}^{T} E_{(s_t,a_t)\rho_\pi} \left[r(s_t, a_t) + \lambda H \left(\pi \left(\hat{A} \cdot \middle| s_t \right) \right) \right] \tag{11}$$

TABLE 5. Performance comparison (without TP).

	Collision rate	Travel Time of Vehicles (s)	Travel Time of Pedestrian (s)	Total Travel Time (s)
SAC	0.004	6.385	5.769	12.154
Aggressive IDM	0.017	4.121	7.576	11.697
Conservative IDM	0.525	8.29	5.295	13.585

TABLE 6. Performance comparison (with TP).

	Collision rate	Travel Time of Vehicles (s)	Travel Time of Pedestrian (s)	Total Travel Time (s)
SAC	0.004	6.1	5.462	11.562
Aggressive IDM	0.141	6.254	6.119	12.373
Conservative IDM	0.372	8.172	5.101	13.273

where λ is the temperature parameter that controls the relative importance of entropy against reward. In other words, it controls the extent of exploration of the RL policy. where λ is the temperature parameter that controls the relative importance of entropy against reward. In other words, it controls the extent of exploration of the RL policy.

The entropy term could be further expanded as:

$$H\left(\pi\left(\hat{A}\cdot\middle|\, s_t\right)\right) = -E_{a_t}\log\pi\left(a_t|\, s_t\right) \tag{12}$$

With this objective design, the policy is incentivized to remember to explore the action space when maximizing the expected return. Unlike deterministic RL algorithms such as deep deterministic policy gradient (DDPG), the policy here gives similar importance to near-optimal actions that are approximately equally attractive, which encourages the exploration of the action space so that the agent is less likely to get stuck in local optimums.

There are two models (i.e., critic and actor) that need to be trained for SAC: $Q_\theta(s_t, a_t)$ and $\pi_\phi(a_t|s_t)$. Neural networks are adopted for the two models. The Q-function $Q_\theta(s_t, a_t)$ is for the critic, which takes (s_t, a_t) pair at time t and outputs its estimated long-term expected reward. The tractable policy $\pi_\phi(a_t|s_t)$ parameterized by ϕ takes state at time t and produces probabilities for each action. In this study, a multi-layer perceptron (MLP) with 2 hidden layers (each has 64 neurons) is adopted for both the critic and the actor models. In addition, since the action space (e.g., vehicle acceleration) is continuous, the output of the policy network is a Gaussian distribution with mean and covariance. For actions in the vicinity of the optimal action, their probabilities are similar to the optimal action's probabilities but lower.

The parameters of the networks were updated based on the Adam optimization algorithm [16]. The training of SAC behaves stably and is insensitive to different random seeds. It converges faster than traditional RL methods, encourages exploration, and ultimately produces higher convergence episode reward. It has been empirically shown that SAC outperforms other RL algorithms such as deterministic policy gradient (DDPG) and proximal policy optimization (PPO) for continuous control problems [20].

E. TRAINING SETUP AND HYPERPARAMETERS

The code was run on a computer with a Windows 11 operating system configured with RTX3060 and an Intel(R) Core (TM) 1.6-GHz Intel processor with 16 GB of RAM.

During the model training and evaluation, the RL agents collect information from the Carla simulation platform to identify the system state, calculate the reward, and send driving instructions to Carla for the ego vehicle. Some important hyperparameters used in training include discount factor $\gamma = 0.99$, learning rate =0.0003, and training batch size = 64. The size of the memory replay buffer is set to 50,000 [20]. The temperature coefficient λ in the objective function of SAC is automatically learned through a dual objective [17].

F. RESULTS AND DISCUSSION

The network was trained for two types of pedestrians with/without TP scenarios with random initial pedestrian-vehicle distances (uniformed from 20 m to 40 m), fixed initial vehicle speed (40 km/h), and pedestrian locations. The training was repeated for 5000 episodes, with one episode meaning a pedestrian-vehicle interaction event.

Figure 7 shows the changing of rolling mean episode reward with respect to the training episode. The rolling mean episode reward is the average of mean episode rewards across a rolling window with size 100. Multiple runs of training were conducted and the results were aggregated: solid-colored lines represent the mean across multiple runs and shaded areas represent the mean ± standard deviation interval.

As can be seen, SAC begins to converge from approximately the 800th episode when facing the two types of pedestrians. However, when faced with pedestrians under TP, SAC exhibited more aggressive interaction behaviors during the exploration process, reflected by the drastic changes in rewards in the early stages After. convergence, pedestrians

and vehicles with TP perform slightly better in pedestrian - vehicle interaction than those without TP. That further confirms the necessity of considering pedestrian heterogeneity [15].

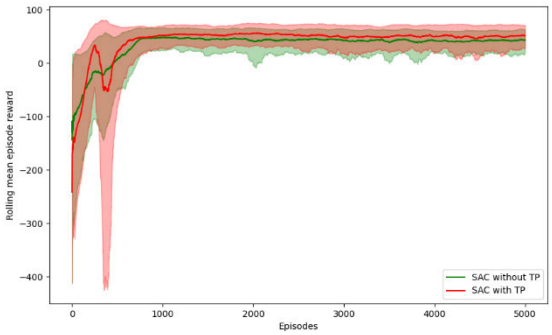


FIGURE 7. Changing of reward during training.

For the evaluation, we tested the average performance of SAC and two other baselines under different TP conditions. The two baselines are the aggressive and conservative reactive-control models, based on the IDM model [21], [22]. The aggressive IDM won't decelerate until the pedestrian starts to cross the street, while the conservative IDM treats pedestrians as obstacles from the beginning and adopts a mild deceleration behavior. For each scenario, we run for 1000 times and calculated the average collision rate, average travel time of vehicle and pedestrian, and the average total travel time.

Table 5 and Table 6 shows the performance of the tested models. When facing pedestrians without TP, radical IDM and conservative IDM respectively obtained the travel time of vehicles and pedestrians, representing selfishness and altruism, as well as the two extremes of efficiency and safety. At the same time, the SAC model performs between the two baselines in terms of efficiency. On the other hand, when facing TP, SAC performs better than the two baselines. The results indicate that different strategies could be adopted when facing heterogeneous pedestrians, such as accelerating the reduction of pedestrian crossing intention when facing pedestrians without TP and making concessions when facing pedestrians with time pressure to avoid conflicts.

Overall, the SAC guarantees safety, while both baselines result in collision. It is worth noting that the conservative model has a higher collision rate than the aggressive model. This is because the dangerous initial conditions lead to the case that most of the time, pedestrians inevitably collide once they cross the street. At the same time, the deceleration amplitude adopted by the model misleads pedestrians, causing them to mistakenly believe that vehicles are giving way and crossing the street.

To further explain the strategies learned by SAC, we extracted several examples, including PVI events at the initial distances of 20, 30, and 40 meters. Figure 8 and Figure 9 show the pedestrian-vehicle distance (defined as the

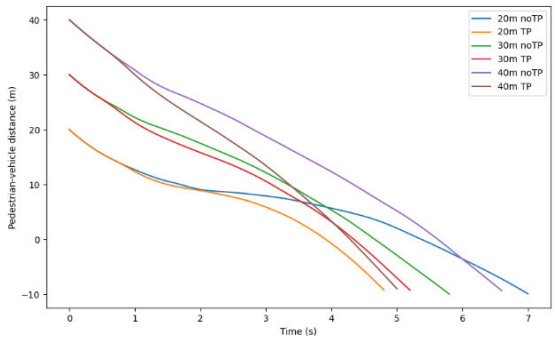


FIGURE 8. Distance changes in a single PVI event.

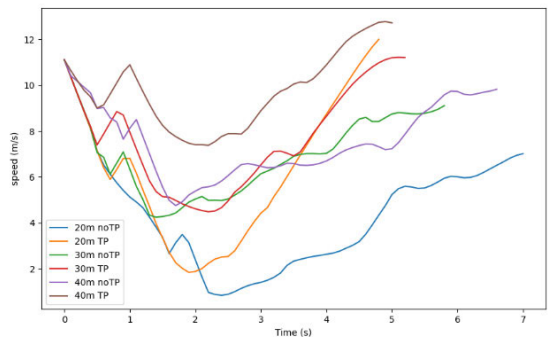


FIGURE 9. Speed changes in a single PVI event.

coordinate difference between pedestrians and vehicles in the direction of vehicle movement) and vehicle speed during the PVI process under different initial conditions, respectively. The SAC learned a conservative strategy of adopting different amplitudes of deceleration based on different initial conditions, and then it chose to speed up after confirming that pedestrians would not cross or the pedestrian crossing had been completed.

It is a topic worth further discussing how to encourage multi-mode driving strategies, such as accelerating from the beginning to force pedestrians to reduce their intention to cross. Although aggressive driving strategies may have a lower collision rate, the inherent uncertainty of pedestrians hinders the exploration of aggressive strategies by vehicles, as acceleration strategies always sacrifice safety.

V. CONCLUSION

Combining game theory and reinforcement learning, this study proposes a novel method to ensure the efficiency and safety of speed decisions for AVs' interaction with heterogeneous pedestrians on unstructured roads. We conducted a pedestrian crossing experiment in a VR platform. The extended PVG model was constructed based on the experimental data, considering the pedestrian heterogeneity. After the training of SAC, AVs can interact well with pedestrians under different conditions on the roadside of unstructured roads. The results validated the effectiveness of the PVG model in capturing pedestrian crossing intentions and the

necessity of considering pedestrian heterogeneity for speed decision-making. The proposed model contributes to improving public acceptance of autonomous driving.

However, this study has certain limitations. The behavioral data collected in the VR environment may not fully replicate real-world pedestrian dynamics and potential discrepancies in risk perception between virtual and physical environments. While the VR-based experiment provides controlled conditions for analyzing pedestrian decision-making, its generalizability to real-world traffic requires further validation through field studies or hybrid simulation platforms. In future work, this study can further be improved by designing a better RL algorithm and considering more actions and comprehensive indicators of the PVI process. Specifically, RL algorithms can enhance the learning of sparse rewards. A prediction and decision-making model that integrates pedestrian heterogeneity could be designed. Moreover, extending the framework to multi-agent scenarios, where multiple pedestrians interact simultaneously with AVs, is critical for addressing complex urban environments. And more reasonable social rewards need to be designed to encourage vehicles to learn aggressive driving strategies, balancing safety and efficiency.

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